AMLAL EL MAHROUSS

Boost.Math and its Constants. Why?

Apery’s Constant.

$$\sum_{n=1}^{\infty} \frac{1}{n^3} = \zeta(3).$$

Some Notes:

- The Apery's constant is irrational.
- However, it has not proven to be transcendental.
- But it has known to be an algebraic period.

$$\sum_{n=1}^{\infty} \frac{1}{n^3} = \zeta(3).$$

$$\alpha = \int_{P(x_1,\ldots,x_n)\geq 0} Q(x_1,\ldots,x_n)\,\mathrm{d}x_1\ldots\mathrm{d}x_n$$

References: https://mathworld.wolfram.com/AperysConstant.html, https://oeis.org/A002117, https://en.wikipedia.org/wiki/Period_(number_theory)

Reference:
https://github.com/boostorg/math/blob/develop/include/boost/math/constants/constants.hpp

Boost.Math’ constants.hpp

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BOOST_DEFINE_MATH_CONSTANT(zeta_three, 1.202056903159594285399738161511449990e+00, "1.2020569031595942853997381615114499907649862923404988817922715553418382057863130901864558736093352581461991577 80e+00")

Why?

- Performance Reasons.
- Much cleaner to write as a macro.
- https://oeis.org/A002117